BIRZEIT UNIVERSITY

Electrical and Computer Engineering
ENCS5325 - Wireless Sensor Networks and Internet of Things – summer 2025
Assignment: Python + Packet Tracer

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1. Introduction

This project is developed with Cisco Packet Tracer and Python to model a smart fire detection and response system. Sensors (smoke detectors, CO_2 sensors, temperature sensors, sprinklers, and alarm sirens) are connected in the Packet Tracer to a home gateway. Rules are implemented for automatically enabling or disabling alarms and sprinklers based on sensor input (smoke, CO_2 level, or temperature).

Besides the simulation, Python scripts (gatewayClient.py and cloudServer.py) are used to simulate a gateway–cloud network. The client transmits sensor events and their IDs (we take this ID: 1222177) to the cloud server, and the latter receives and logs them.

The Packet Tracer instructions and Python scripts collectively demonstrate how IoT systems enable automation, monitoring, and remote processing of data in a smart environment.

2. Objective

It aims to emulate and develop a smart fire protection system that can independently sense risk and react in real-time. It uses Cisco Packet Tracer to network IoT devices and a home gateway to control sprinkler, alarm, and other actuators. Python programming is applied to facilitate simulation by handling gateway–cloud communication, presenting how sensor data can be remotely accessed.

3. System Design

It is built with two layered integrated systems. In Cisco Packet Tracer, smoke detectors, CO_2 sensors, and temperature sensors are all connected to a Home Gateway and actuators such as sprinklers and alarms, powered by smart rules (e.g., when smoke intensity hits a certain level, the sprinkler and alarm are activated). Augmenting the simulation, Python communication layer allows for data transfer between gateway and cloud, and gatewayClient.py posts sensor events with ID 1222177 while cloudServer.py reads and logs the events to illustrate how cloud monitoring can enable real-time fire safety response.

IP Addressing:

Device	Interface	IP Address	Subnet Mask	Default Gateway
Home Gateway (DLC100)	LAN (Ethernet)	192.168.0.1	255.255.255.0	—
Server-PT	Fa0	192.168.0.2	255.255.255.0	192.168.0.1
IoT Devices (DHCP)	Wi-Fi / LAN	192.168.0.100 → 192.168.0.200	255.255.255.0	192.168.0.1
Actuators (Fan, Siren)	Wi-Fi / LAN	Assigned by DHCP	255.255.255.0	192.168.0.1

Table 1 IP addressing

4. Block Diagram and System Functionality

5.1 Smoke Sensor

A smoke sensor detects the presence of smoke particles in the air. It is the primary component for fire detection and sends an immediate signal to the home gateway when smoke levels rise above a threshold.

Figure 1: Smoke Sensor

5.2 CO_2 Sensor

The CO_2 sensor continuously monitors the concentration of carbon dioxide in the environment. High CO_2 levels can indicate poor air quality or potential hazards, allowing the system to raise early warnings even before visible smoke appears.

Figure 2 : CO_2 Sensor

.Fire/Flame Sensor

A flame sensor detects infrared radiation emitted by flames or high heat sources. It provides rapid confirmation of fire events, reducing false alarms and ensuring faster response.

5.4 Home Gateway (DLC100)

The home gateway acts as the central hub of the system. It wirelessly connects to

sensors, manages their status, and applies predefined rules. It also links the local devices to the server and cloud layer for monitoring and control.

5.5 Server-PT

The server functions as the IoT registration and control platform. It stores sensor data, manages devices, and hosts a dashboard for real-time monitoring. It can also run additional scripts (e.g., Python) for intelligent decision-making and logging.

5.6 **Actuators (Siren, Fan, Sprinkler)**

5.6.1 Siren:

Emits a loud alarm to alert occupants of danger.

5.6.2 Fan:

Activates during warning states to reduce smoke concentration and improve air circulation.

5.6.3 Sprinkler:

Automatically releases water to suppress flames in case of confirmed fire.

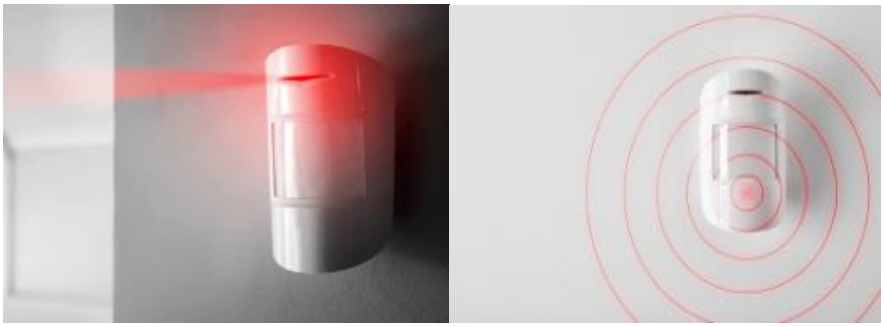
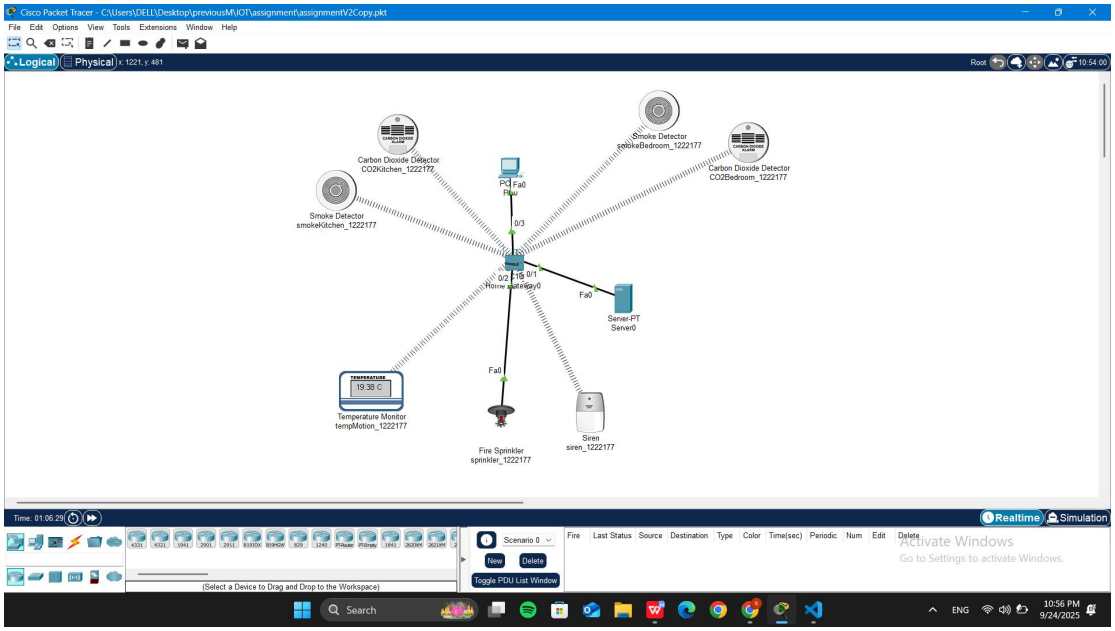


Figure 3: Siren

5.7 Cloud Layer

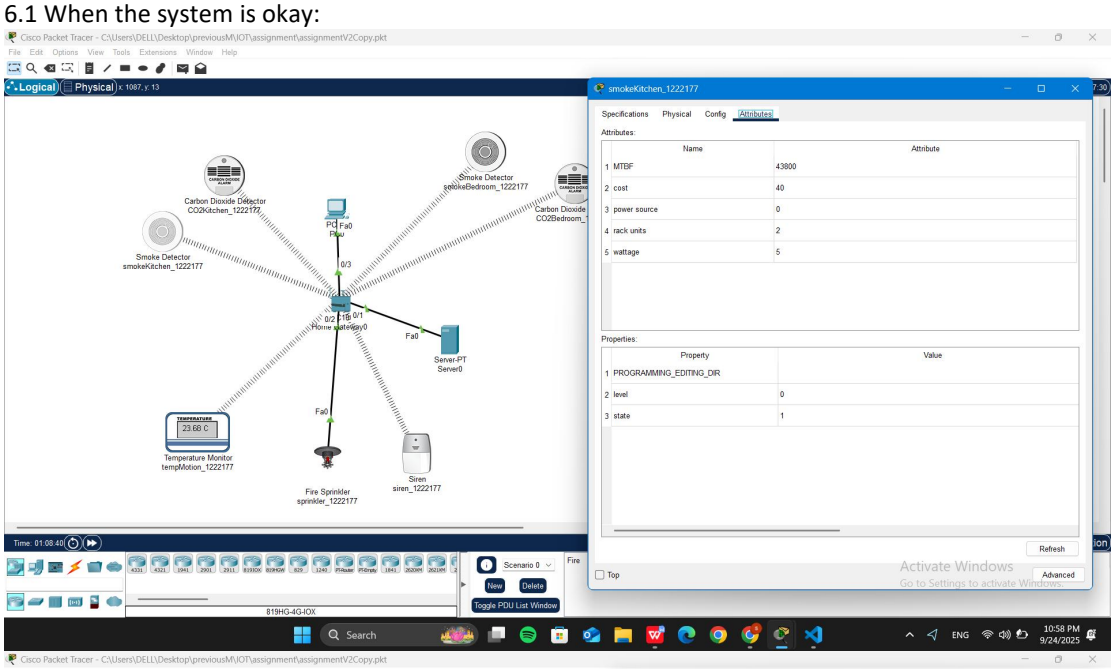
The cloud layer connects the home gateway to external services. It forwards alarms and updates to the Fire Management Department and hosts a centralized dashboard that provides real-time visualization of system status and events.

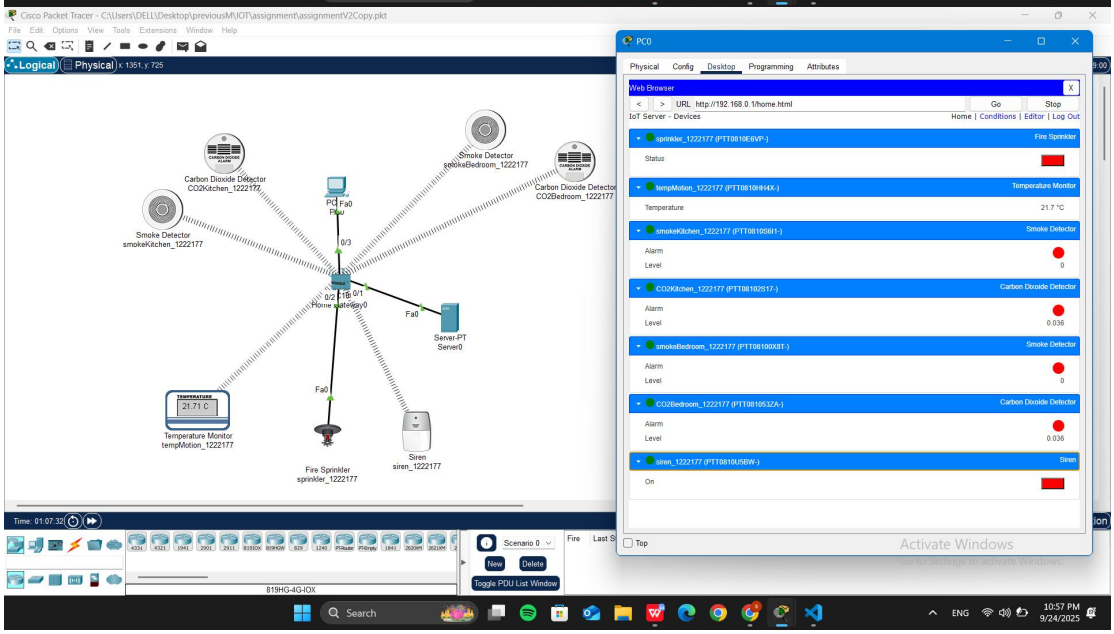
5. Toplogy and code Implementation :



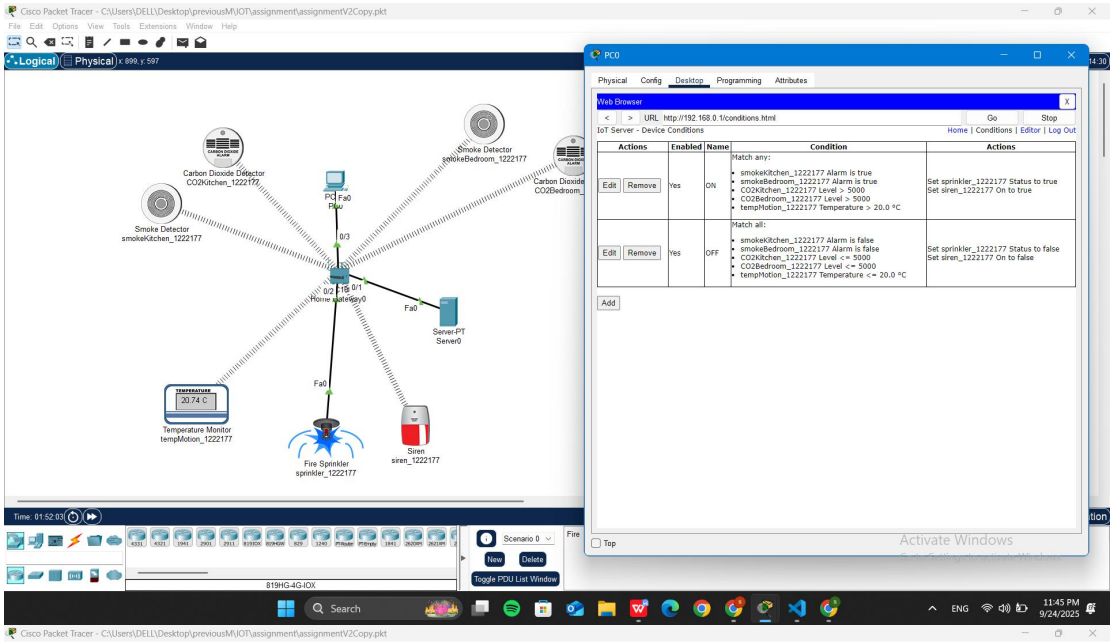
6. Testing & Results:

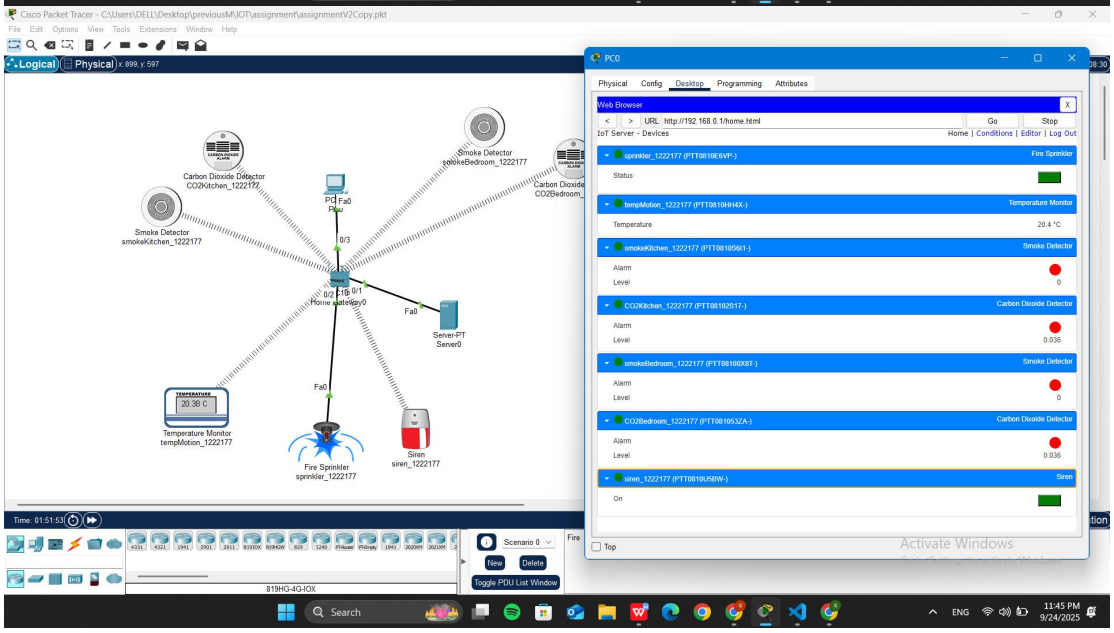
6.1 When the system is okay:

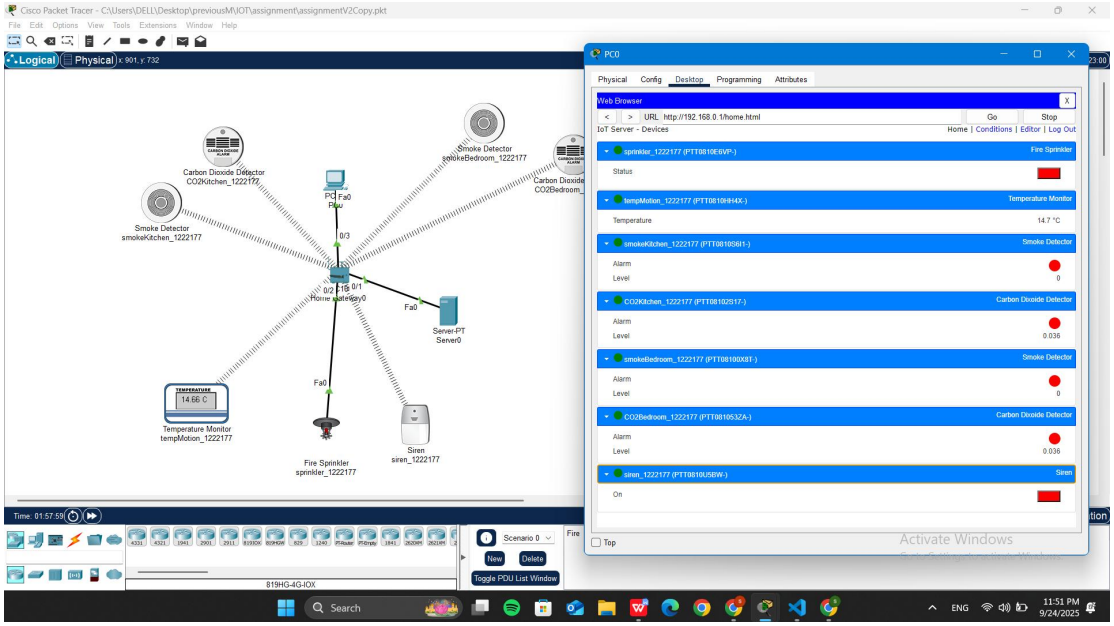




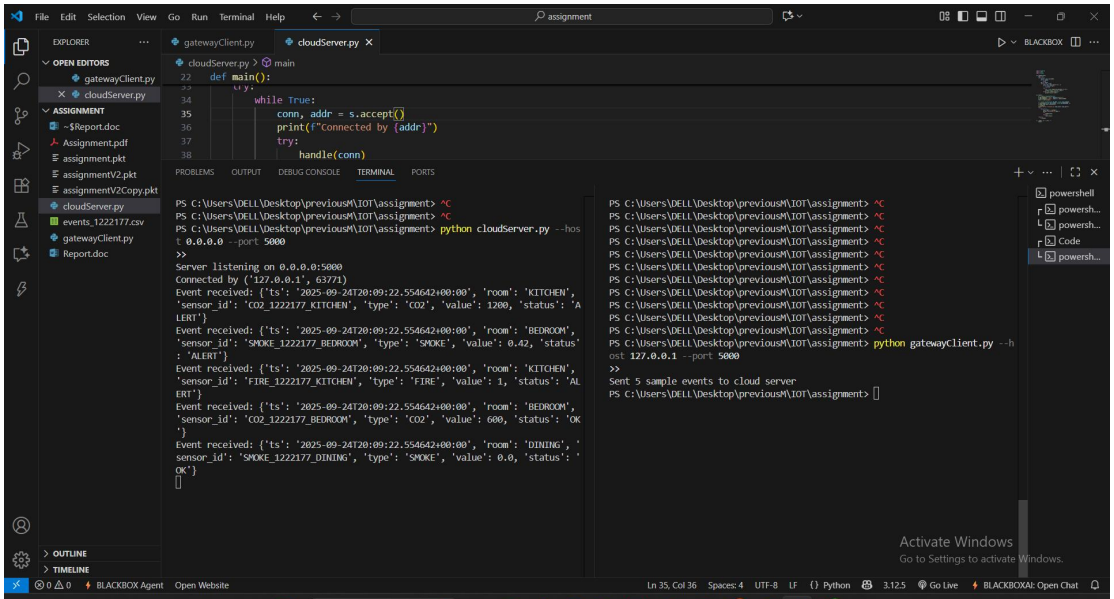
6.2 when the system is not okay

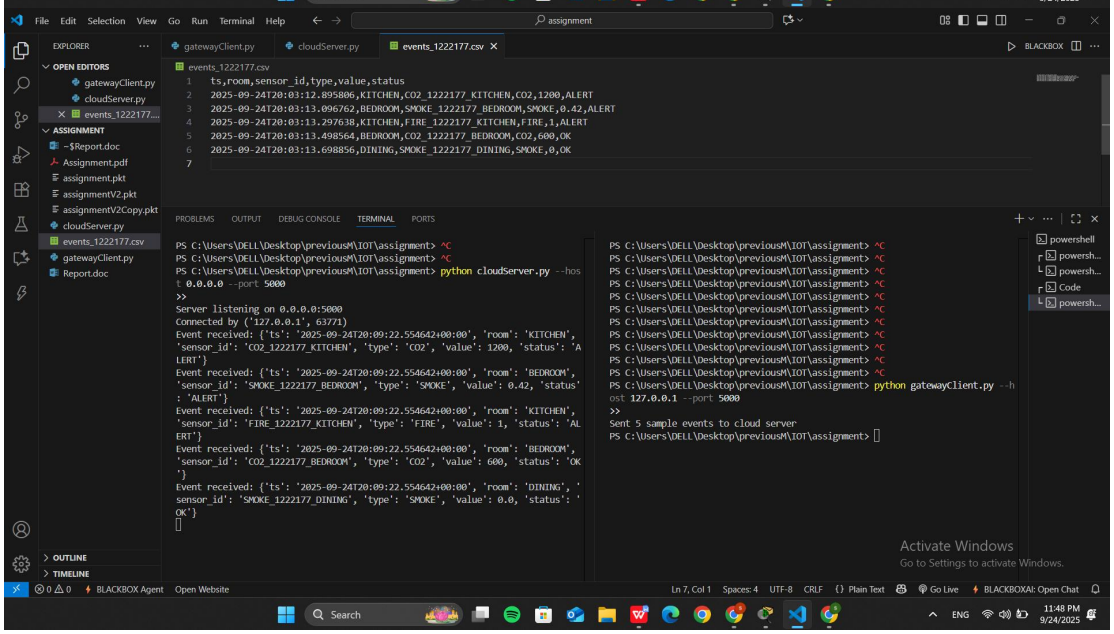





Cisco Packet Tracer - C:\Users\DELL\Desktop\previousM\IOT\assignment\assignmentV2Copy.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 901, y: 732
Carbon Dioxide Detector
CO2Kitchen_1222177
Smoke Detector
smokeKitchen_1222177
PC0
Fa0
0/3
0/2
0/1
Fa0
Server-PT
Server0
Fa0
Temperature Monitor
tempMotion_1222177
14.66 C
Fire Sprinkler
sprinkler_1222177
Siren
siren_1222177
Smoke Detector
smokeBedroom_1222177
Time: 01:57:59
819HG-4G-IOX
Scenario 0
New
Delete
Toggle PDU List Window
PC0
Physical
Config
Desktop
Programming
Attributes
Web Browser
URL http://192.168.0.1/home.html
Go
Stop
IoT Server - Devices
Home | Conditions | Editor | Log Out
sprinkler_1222177 (PTT0810E6VP-)
Fire Sprinkler
Status
tempMotion_1222177 (PTT0810HH4X-)
Temperature Monitor
Temperature
14.7 °C
smokeKitchen_1222177 (PTT081006I1-)
Smoke Detector
Alarm
Level
0
CO2Kitchen_1222177 (PTT08102517-)
Carbon Dioxide Detector
Alarm
Level
0.036
smokeBedroom_1222177 (PTT08100X81-)
Smoke Detector
Alarm
Level
0
CO2Bedroom_1222177 (PTT081053ZA-)
Carbon Dioxide Detector
Alarm
Level
0.036
siren_1222177 (PTT0810U5BW-)
Siren
On
Top
Activate Windows
Search
ENG
11:51 PM
9/24/2025

6.3 the code result:





7. Conclusion:

This project successfully integrated IoT concepts into an actual smoke detection and fire alarm system. By bringing together Packet Tracer for the edge layer and Python for the cloud layer, we demonstrated how sensors, actuators, and gateways could be utilized together to facilitate safety and automation. The system is capable of detecting smoke, high levels of CO_2, and fire, and then triggering alarms and sprinklers and sending notifications to the cloud for monitoring and visualization. This technique highlights the potential of IoT in real-life applications of safety where response time and remote monitoring are equally important. The project emphasized appropriate system design, harmonious communication between layers, and rules for automation to ensure reliability. Overall, it demonstrates how IoT can enhance safety, efficiency, and awareness.